Electric Power Components and Systems

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Submission ID	245596946
Article Type	Research Article
Keywords	Electric vehicle, Incremental capacity curve, lithium-ion battery, Multi-health indicators, Prognostics and health management, , Remaining useful life prediction
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Accurate prediction of lithium-ion battery's remaining useful life using deep learning models with correlation-based novel multi-health indicators

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Abstract

Accurate online estimation of capacity and lifespan forecasting of lithium-ion batteries (LIBs) is crucial for adopting electric vehicles. Data-driven strategies, particularly those incorporating battery health indices (HIs) and deep learning methods, are increasingly popular in this field. However, deriving effective HIs for predicting a battery's lifespan remains challenging. Previous research has often focused on battery characteristics during the constant current (CC) charging phase, neglecting the discharging phase's impact on battery aging. This paper introduces a framework for predicting remaining useful life (RUL) using novel HIs and stacked long short-term memory (SLSTM). The method leverages the optimized incremental capacity (IC) curve properties during the CC-charging and CC-discharging phases. Pearson and Spearman correlation analyses indicate a significant correlation between the IC curve and battery capacity during CC-discharging, affecting the overall lifespan. Initially, the IC curve derives the health index from the charging and discharging phases. Cycle and capacity are also analyzed as additional health indices. These multidimensional indices are input into the SLSTM model to forecast the LIB's RUL. Comparative analysis with models such as gated recurrent unit (GRU), attention-based SLSTM, and GRU models, as well as recent work using the NASA AMES battery dataset, demonstrates the proposed models' superior performance.

Keywords: Electric vehicle, Incremental capacity curve, lithium-ion battery, Multi-health indicators, Prognostics and health management, Remaining useful life prediction.

1 Introduction

The contribution of fossil fuel-based transport to air pollution and greenhouse gas emissions has received considerable attention [1, 2]. Environmentally friendly technologies have advanced significantly and promise to lower greenhouse gas emissions and improve air quality [3, 4]. Lithium-ion batteries find extensive application in emerging technologies due to their impressive energy density and extended operational life [5–8]. These batteries (LIBs) represent the optimal choice for energy storage for EVs. Their secure and pragmatic use as a power source enhances their overall worth and encourages secondary utilization and the recycling of materials. Simultaneously, this reduces the carbon footprint during battery production and recycling phases [9, 10]. As a result, the dependability and safety of the LIBs have received attention.

Nonetheless, it's essential to acknowledge that the decline in the performance of LIBs is an inevitable process [11]. Conventionally, lithium-ion batteries are considered to have reached the end of their operational life when their capacity drops to approximately 70% of their initial value or when the internal resistance doubles. Even at this stage, these batteries can still serve in various capacities, such as energy storage systems and electric bicycles [12, 13]. However, unlike controlled laboratory conditions, real-world usage patterns can lead to erratic battery performance and capacity degradation. This unpredictability may lead to the inefficient use of battery cells, resource wastage, and the potential for safety hazards when there's a deficiency in precise prediction of the RUL. A comprehensive understanding of the evolving degradation patterns in battery lifespan is crucial to avoiding premature replacements and excessive usage. Consequently, predicting the lifespan and estimating the capacity of lithium-ion batteries face substantial challenges [14].

Predicting the RUL of LIBs typically involves model-driven and data-driven approaches. In the model-driven system, the inherent aging process of batteries is leveraged to create intricate mathematical formulations that capture their responses. These models are meticulously crafted with mechanistic knowledge developed through mathematical deduction. Examples of mathematical techniques used to parameterize these models include the expectation-maximization method [15] and the extended Kalman filtering method [16], among others. However, achieving precise forecasts is challenging due to the intricacy of the battery aging process and potential disparities between data dispersion and the underlying framework hypothesis.

The data-driven approach has garnered significant courtesy recently due to its accessibility to data and the advancements in machine learning techniques. Machine learning, artificial neural networks, and deep learning are considered data-driven methods for forecasting RUL. Data-driven RUL forecasting typically follows two main approaches. One approach directly uses the data to make RUL predictions. In the other approach, the data is initially preprocessed to extract relevant health indicators, and then appropriate models are applied to these indicators for RUL prediction.

Various direct data-driven methods, including traditional machine learning models like GBDT [17], Gaussian regression [18], random forest [19], and deep learning techniques like MLP [20], RNN [21], LSTM [8], Dual-LSTM [22], GRU [23], Transformers [24], and so on, are being widely employed as direct methods for forecasting the RUL of Lithium-ion batteries. These methods offer different advantages and trade-offs regarding computational efficiency and their ability to capture temporal information and long-range dependencies. Transformers, known for their effectiveness in encoding sequences, are gaining attention for their potential to enhance RUL prediction alongside traditional physical models.

Numerous research studies have delved into predicting the RUL of batteries through data-driven methods involving the extraction of health HIs from diverse sources, followed by training predictive models using these indicators. These research studies employ a range of health indicators and methodologies. These indicators include input voltage, current, and temperature measurements and factors like incremental capacity curves, charge-discharge periods, isobaric voltage drop time, and many more [18, 19, 25–31]. In a most recent study by Luo et al. [32], AutoML [33] techniques were used to predict the RUL of LIBs. HIs were extracted using incremental capacity (IC) analysis, and a smoothing IC curve based on the Kalman filter algorithm was used. Three HIs were identified as effective in characterising the ageing phenomenon throughout the charging process, including the constant current (CC) and constant voltage (CV) phases. Nonetheless, the approach presented in [32] necessitates extended training time, focusing exclusively on the CC-CV charge cycle while overlooking the discharge cycle data when deriving the health indicators. This limitation makes it less suited for swift and dynamic RUL prediction.

Moreover, recent literature surveys indicate that the aging pattern of lithium batteries strongly correlates with the peaks observed in the IC curve of both the charging and discharging cycles [34][35][36]. Additionally, features extracted from the CC discharge phase, including voltage degradation rate and capacity fade, are crucial for detecting battery degradation signs and indicating ageing-related phenomena impacting battery RUL. Constant current discharge supports controlled experiments simplifies data collection and enhances replicability. Its simple profiles ensure safety and efficiency in battery characterization and predictive modelling. Thus, we employ an optimal, streamlined feature extraction technique based on IC curves to estimate RUL from historical data. This technique efficiently extracts features from the CC charge and discharge phases, which correlate highly with RUL. Remarkably, this approach has not been previously utilized for this purpose.

The significant contributions of this method include:

i. The peak and position values from the IC curve during the CC phase of charging and the CC phase of discharge cycles are extracted as health indicators. Additionally, the number of cycles and the battery's capacity are direct health indicators.
ii. A strong and dependable correlation between these HIs and the battery's capacity is assessed through Pearson and Spearman correlation analyses.
iii. A preprocessed dataset, including HIs, battery capacity, and the corresponding number of cycles, is used to train an LSTM-based model (SLSTM) for predicting the remaining useful life (RUL) trends of lithium-ion batteries (LIBs). We also analyze benchmark models GRU, attention-based SLSTM, GRU models within the same context for comparative evaluation.
iv. The proposed method's performance has been validated in three scenarios: for individual batteries, in the 2P1 configuration, and using a leave-one-out strategy.
v. The method's accuracy, effectiveness, robustness, and superiority over existing traditional data-driven approaches have been verified by the NASA Lithium-ion battery dataset. Furthermore, the proposed method can accurately predict the estimated RUL of the EV LIBs with an MAE and RMSE of 0.0589 and 0.0730, respectively.

Table 1 displays relevant notations and definitions used throughout the paper.

Table 1: Notations with their definitions.

Notations	Definitions
EV, RUL, HIs	Electric vehicle, Remaining useful life prediction, Health- indicators, respectively
CC, CV	Constant current and Constant voltage phase, respectively
$LIBs, IC$	Lithium-ion Batteries, Incremental capacity curve, respectively
$SLSTMP, ASLSTM$	Stacked long short-term memory, Attention-based SLSTM, respectively
$GRU, AGRU$	Gated recurrent unit and Attention-based GRU, respectively
ICA, SOH	Incremental capacity analysis, State oh health, respectively
TCN, DNN	Temporal convolutional network, deep Neural network, respectively
$ARIMA, LSTM$	Autoregressive integrated moving average, Long short-term memory, respectively

The paper's structure is as follows: Section II offers a concise review of previous research on the RUL prediction of lithium-ion batteries. Section III comprehensively explores problem formulation and an in-depth analysis of deep learning models, including LSTM, SLSTM, and GRU. Section IV provides an extensive account of our proposed model, encompassing the architecture of the methods, data sources, HIs handling, correlation analysis, and the training process. Section VI encompasses performance metrics and a thorough presentation of experimental results. Lastly, in Section VII, we draw conclusions based on the findings.

2 Related work

A data-driven framework utilizes historical data to predict battery degradation patterns. Weight parameters within this framework are established through training data analysis without requiring physics-based techniques. Data-driven approaches are gaining global recognition for their adaptability and broad applicability. Data-driven RUL prediction follows two main approaches. In one approach, data is directly used for RUL predictions. In the other approach, the data is initially preprocessed to extract relevant health indicators, and then suitable models are applied to these indicators for RUL prediction.

The authors in [37] proposed a Gaussian process regression algorithm for predicting the remaining useful life of lithium-ion batteries (LIBs), demonstrating excellent long-term forecasting performance. The authors in [38] explored a hybrid prognostic approach for RUL estimation in LIBs. This approach integrated the grey model, autoregressive integrated moving average, and variational mode decomposition models, resulting in a comprehensive RUL prediction strategy. The authors in [39] introduced an innovative method for estimating the state of health (SOH) and predicting RUL in LIBs. Their approach leveraged the IC curve during the constant current discharge phase and was based on ANN with four layers. Authors in [40] proposed a cloud-edge collaborative technique for estimating the SOH in LIBs and optimizing capacity prediction. Their method incorporated a backpropagation neural network enhanced by a genetic algorithm.

Researchers have recently explored various methods based on deep learning for predicting different systems' RUL. Dong et al. [41] used LSTM to tackle the gradient explosion problem during RUL prediction. Zraibi et al. [42] introduced a comprehensive CNN-LSTM-DNN algorithm, where each hybrid approach played a crucial role in enhancing RUL prediction. Wang and associates [43] proposed an innovative hybrid method for online life prediction, combining a BiLSTM-AM model with support vector regression (SVR). This method involved gathering initial data, updating them with SVR, and using BiLSTM-AM to estimate cycle life. Similarly, authors in [44] fuse ARIMA-LSTM models to predict LIBs' RUL, exploiting both linear and nonlinear battery degradation patterns. In another approach, the author in [45] employed a self-designed LSTM network called IRes2Net-BiGRU-FC and DNN to predict both high-frequency and low-frequency parts of the original data, demonstrating robust RUL prediction during both the CC and CV phases. Recently, Li et al. [46] utilized a hybrid TCN-GRU-DNN model with dual attention mechanisms, effectively capturing battery regeneration and decay trends while enhancing prediction accuracy and efficiency. Similarly, Yang et al. [47] employed TCN, GRU, and attention mechanisms. Ge et al. [48] developed an RUL prediction model for lithium-ion batteries using a multi-layer convolutional network and multi-head attention mechanism. Lee et al. [49] proposed a hybrid TCN-XGBoost model to explore batteries' nonlinear and evolving characteristics.

Data-driven RUL prediction based on HIs encompasses the extraction of these HIs and subsequent model training. Numerous scholars in this domain have conducted extensive research. Ansari et al. [25] explored RUL prediction through health indicators. Wang et al. [26] considered inputs such as voltage, current, and temperature measurements as HIs for RUL prediction, utilizing the Gradient Boosting Decision Tree. Meanwhile, Meng et al. [18] and Wang et al. [27] employed a multi-HI approach, extracting data from incremental capacity curves and charge-discharge periods, utilizing the Box-Cox transformation method to construct a battery state of health estimation framework. Zhang et al. [28] utilized isobaric voltage drop time as an HI, comparing various neural network models for battery RUL prediction. Li et al. [29] derived the open circuit voltage (OCV) from current pulses as an HI, deploying support vector regression optimized by the swarm algorithm to address nonlinear data processing and convergence accuracy. Yang et al. [19] considered capacity degradation and internal resistance growth as HIs for battery SOH estimation. Liu et al. [30] regarded battery OCV, current, and impedance as HIs, employing adaptive Brownian motion for SOH estimation. Furthermore, Liu et al. [31] utilized Light Gradient Boosting Machines (LightGBM) with a range of HIs, including electrochemical impedance spectroscopy, OCV, temperature, and incremental capacity-differential voltage (IC-DV) curves for RUL prediction. An automated machine learning (AutoML) model [32] has recently been applied to the RUL prediction of LIBs, with HIs extracted through IC analysis. The authors have comprehensively analysed the CC and CV phases, highlighting the strong correlation between the IC curve during the CC phase and the charging capacity in the CV phase and their impact on battery lifespan.

Moreover, recent studies have explored both intra-battery (training and testing on similar batteries) [25, 50] and inter-battery (training on one set of batteries and testing on another) transfer learning methods [51, 52] for various Lithium-ion battery datasets with similar or diverse charge-discharge and temperature conditions. Ma et al. [53] investigated multiple transfer learning approaches. Additionally, Wang et al. [54] proposed an optimized local tangent space alignment algorithm to enhance RUL estimation by extracting precise health indicators from charge data.

In recent years, the estimation of lithium-ion battery SOH or RUL through incremental capacity analysis (ICA) has garnered significant attention among researchers. In their research, authors in [55, 56] have convincingly demonstrated the close correlation between the peak position and intensity of the incremental capacity (IC) curve during constant current charging processes and the deterioration of LIB SOH. In light of this requirement and acknowledging the constraints of relying on a single health indicator for improving prediction accuracy, the central objective of this paper is to harness the potential of ICA by incorporating multiple health indicators extracted from both the CC charging and CC discharging phases, as well as extra HIs obtained through direct analysis, to improve prediction accuracy significantly.

3 Background

In this section, we outline the problem formulation and introduce the deep learning model employed for our analysis.

3.1 Problem formulation

A precise and timely RUL predictor is vital in providing advance notice of potential EV battery risks. State of Health (SOH), a key health indicator for battery aging, reflects the battery's condition across various charge-discharge cycles, as highlighted in [57]. The subsequent capacity percentage determines RUL:

$$SOH(t) = \frac{C_t}{C_r}\% \tag{1}$$

C_r represents the rated capacity, and C_t signifies the measured capacity for cycle t. As a battery undergoes multiple charge-discharge cycles, its capacity degrades over time. Furthermore, in the context of a battery, end-of-life is closely linked to its capacity. It is typically defined as the point at which the remaining capacity reaches 70-80% of the initial capacity [24].

3.2 SLSTM

We employed a stacked long short-term memory (SLSTM) deep learning model to predict the RUL using the HIs extracted from the NASA LIBs dataset. The organization of the used SLSTM is shown in Figure 1. As depicted in Figure 1, the SLSTM architecture is created by stacking multiple LSTM layers. This stacking process equips the model with the capacity to effectively capture intricate patterns and representations within the input data, thereby enabling a smooth flow of information between successive LSTM layers. Additionally, Figure 2 illustrates a general single-cell LSTM for complete understanding.

In LSTM, the cell state is also called long-term memory. This makes it easier for the LSTM cell to store previous intervals. The input, output, and forget gates aid in recalling the prior sequence. The input gate is a save vector since it chooses which input should enter the LSTM cell. The input gate is a sigmoid function that sums the previous cell state and bias. The output gate is a focus vector because it indicates which data may be passed on to the next state. Lastly, the forget gate determines what to retain in the cell state. It does this by multiplying elements of the cell state vector by either 1 or 0 depending on the forget gate's decision. The current cell state will be remembered if the output is 1. Otherwise, it won't be. The inner workings of a single-cell LSTM can be further understood by referring to the following equations, which thoroughly explain how it processes and stores information.

$$\begin{aligned}
o_t &= \sigma(U_o \cdot h_{t-1} + b_o + W_o \cdot X_t) \quad \text{(Output Gate)} \\
i_t &= \sigma(U_i \cdot h_{t-1} + b_i + W_i \cdot X_t) \quad \text{(Input Gate)} \\
f_t &= \sigma(U_f \cdot h_{t-1} + b_f + W_f \cdot X_t) \quad \text{(Forget Gate)} \\
\tilde{c}_t &= \tanh(U_c \cdot h_{t-1} + b_c + W_c \cdot X_t) \quad \text{(Hidden Cell State)} \\
c_t &= f_t \odot c_{t-1} + i_t \odot \tilde{c}_t \quad \text{(Cell State)} \\
h_t &= o_t \odot \tanh(c_t) \quad \text{(Hidden State)}
\end{aligned}$$

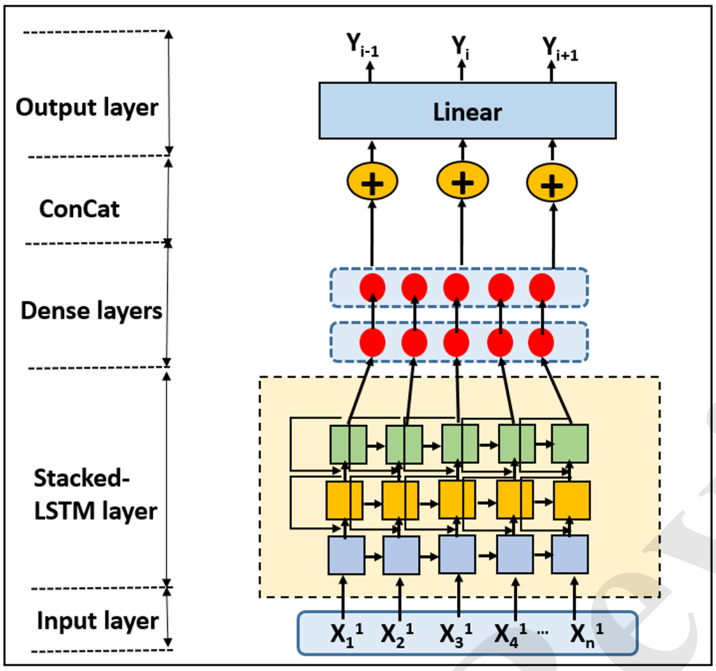


Fig. 1: Stacked LSTM model.

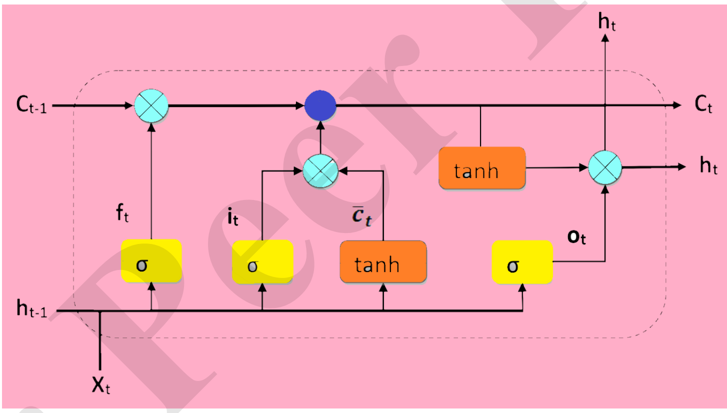


Fig. 2: Single-cell LSTM.

Where the input vector $X_t \in R^d$ represents the input to the cell, while $o_t \in R^h$ represents the activation of the output gate, $f_t \in R^h$ denotes the activation vector for the forget gate, and $i_t \in R^h$ is the activation vector for the input gate. The hidden state vector of the LSTM cell is $h_t \in R^h$, and there's also the intermediate hidden state vector $\bar{c_t} \in R^h$. The cell state vector is denoted as c_t. We use mathematical functions like the sigmoid function σ and the hyperbolic tangent function $tanh$ and operations such as element-wise multiplication ('$\odot$') and standard matrix multiplication ('.'). Additionally, the weight matrix W and the weight matrix U, both in $R^{(h \times d)}$, along with the bias vector $b \in R^h$, are the parameter elements of the LSTM cell.

3.3 ASLSTM

Attention-based stacked LSTM is shown in Figure 3. Here, we add an attention layer to check attention performance in the RUL prediction of the NASA LIBs dataset.

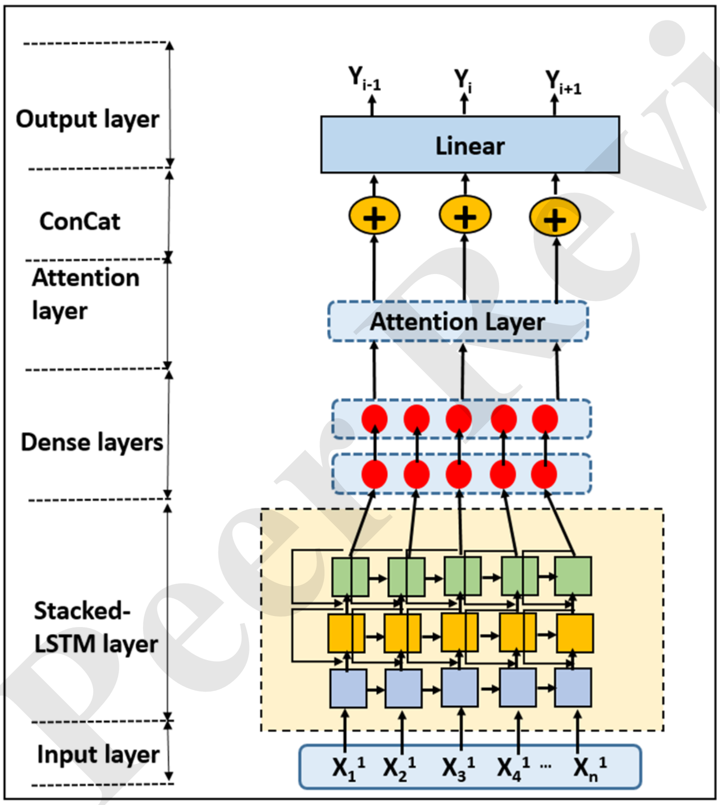


Fig. 3: ASLSTM model.

An attention model, also called an attention mechanism, is a method neural networks use to process input. The neural network's attention mechanism mimics the human brain's visual attention system. The attention framework enables the neural network to selectively concentrate on a small number of pertinent tasks while ignoring others, similar to how humans do not pay attention to every word when reading a text. Usually, the attention framework is coded as a function. Formally, the attention method transforms a series of queries $Q = q_1, q_2, ..., q_m$ and a set of key-value pairs $K, V = (k_1, v_1), (k_2, v_2), .., (k_m, v_m)$ into outputs, where $Q \in R^{m \times d}, (K, V) \in R^{m \times d}$. This context represents the output, keys, query, and values as vectors. The final output matrix is computed as a weighted sum of the values. The weights for each value are determined by a compatibility function that evaluates the relationship between the query and the respective key. The following calculations yield the ultimate final output matrix:

$$Attention(Q, K, V) = softmax\Big(\frac{Q.K^T}{\sqrt{d_k}}\Big)V \quad (2)$$

d_k represents the key and query dimension.

3.4 GRU

The GRU [58] is a variation of Long Short-Term Memory (LSTM) networks, sharing similar functionalities and frequently producing similar results. Figure 4 shows a GRU cell. The GRU cell employs a dual-gate mechanism, distinct from LSTM's inner cell

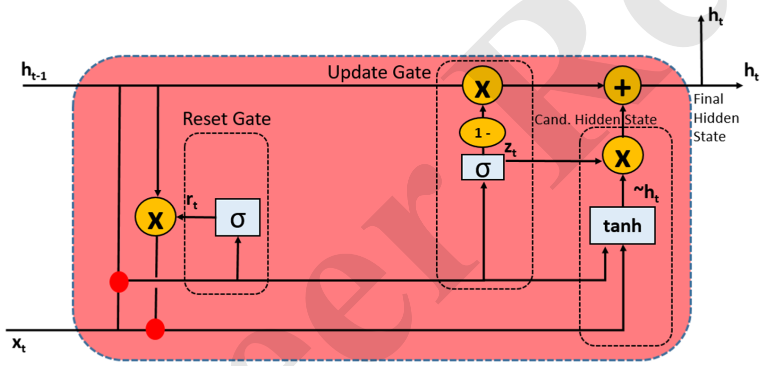


Fig. 4: Single-cell GRU.

state, with one gate dedicated to controlling the transfer of prior information to the next cell, referred to as the update gate. The second gate, the reset gate, determines the degree to which information from the past should be excluded from memory. These gates are mathematically represented as follows:

$$r_t = \sigma(U_r \cdot h_{t-1} + W_r \cdot X_t) \quad \text{(Reset Gate)}$$
$$z_t = \sigma(U_z \cdot h_{t-1} + W_z \cdot X_t) \quad \text{(Update Gate)}$$

$$\tilde{h}_t = \tanh(W \cdot X_t + r_t \odot (U \cdot h_{t-1})) \quad \text{(Candidate Hidden State)}$$
$$h_t = z_t \odot \tilde{h}_t + (1 - z_t) \odot h_{t-1} \quad \text{(Final Hidden State)}$$

The input vector $X_t \in R^d$ acts as the input to the GRU cell, while $z_t \in R^h$ embodies the activation vector for the update gate. Similarly, $r_t \in R^h$ represents the activation vector for the reset gate, and $h_t \in R^h$ stands as the final hidden state vector of the GRU cell. Furthermore, the integral candidate hidden state vector, denoted as $\tilde{h}_t \in R^h$, plays a pivotal role in GRU's computational processes. The sigmoid function is represented by σ, and the hyperbolic tangent function by $tanh$, with '$\odot$' signifying element-wise multiplication and '$\cdot$' indicating standard matrix multiplication. Additionally, the weight matrix $W \in R^{(h \times d)}$, the weight matrix $U \in R^{(h \times d)}$, and the bias vector $b \in R^h$ constitute the weight and bias parameters within the GRU cell.

4 Proposed method

In this section, we have provided a comprehensive overview of the proposed method's architecture, discussed the dataset utilized, detailed the process of extracting multi-health indicators, presented the results of correlation analysis, and explained the model training procedure.

4.1 Architecture

To accurately estimate RUL using historical data, it is essential to employ an optimal feature extraction technique capable of capturing features strongly associated with RUL. The proposed method primarily focuses on enhancing RUL prediction accuracy by extracting critical health indicators responsible for the aging of lithium-ion batteries. The proposed approach leverages multiple health indicators (HIs) obtained through incremental capacity analysis (ICA) and direct analysis. Initially, the IC curve is computed during the CC charging phase, and two HIs, namely max point height and position, are extracted. Subsequently, an IC curve is also generated for the CC discharge phase, yielding the extraction of the other two HIs: max point height and position. These four HIs are combined with the two HIs acquired from direct analysis, namely capacity and cycle, resulting in six HIs. These six HIs comprehensively characterise the battery's aging process during the charging and discharging cycles. Further, a stacked long short-term memory (SLSTM) model is applied to these six HIs to predict RUL. This approach is illustrated in the accompanying Figure 5, visually depicting the process. Figure 5, representing the architecture of the proposed model, primarily comprises two main sections: extracting health indicators (HIs) from charge cycles through the extraction of constant current data and extracting HIs from discharge cycles using CV discharge cycle data. These HIs, in conjunction with the number of cycles and capacity, are then divided into training and testing data sets. The training data is subsequently employed to instruct an LSTM-based model (SLATM), with capacity as the target variable. Following this training phase, the model is utilized to predict the capacity, and its accuracy and efficiency are assessed using the evaluation method discussed in detail in the subsequent sections.

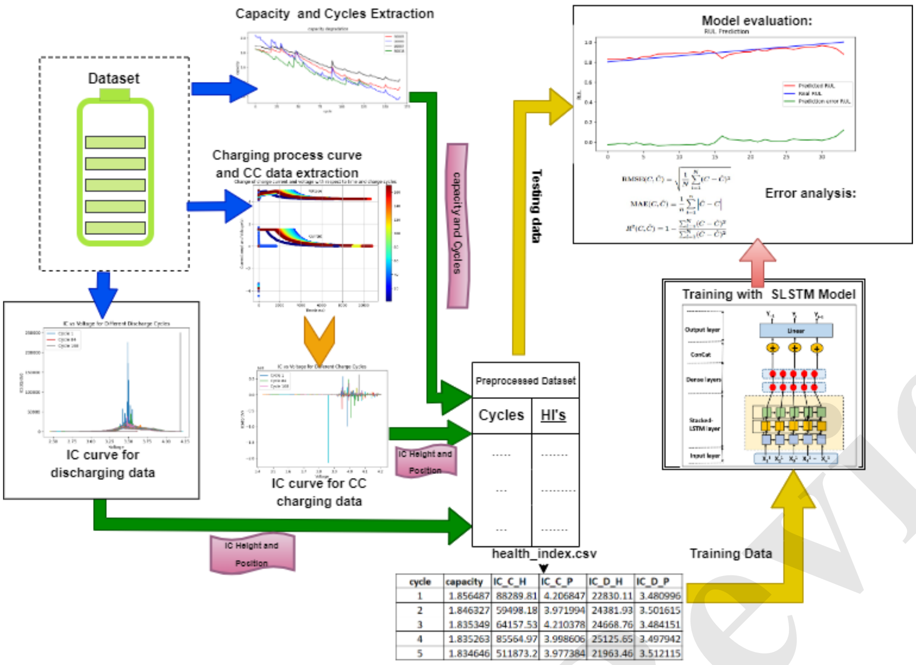


Fig. 5: Architecture of the proposed model.

4.2 Dataset used

The experimental dataset used in this paper was sourced from the NASA AMES Prognostics Center of Excellence and consists of aging data for 18650 lithium-ion batteries (LIBs) [59]. Specifically, data from four LIBs denoted as B005, B006, B007, and B0018 were selected for this experiment. These batteries underwent standard charging and discharging processes at a constant temperature of 25 degrees Celsius. The charging process followed the constant-current (CC) and constant-voltage (CV) principles, i.e. Initially, the voltage was raised to 4.2 V while maintaining a CC of 1.5 A during the CC phase. Subsequently, the voltage was held at 4.2 V under constant voltage conditions until the charge current fell to 20 mA, signifying the CV phase. The batteries were discharged at a constant current of 2 A for discharging. The discharging was carried out until the voltage of each battery, namely B005, B006, B007, and B0018, reached specific cutoff voltages: 2.7 V, 2.5 V, 2.2 V, and 2.5 V, respectively. The experiments concluded when the battery capacity had deteriorated to 70% of its initial capacity. The number of charge-discharge cycles conducted for batteries B005, B006, B007, and B0018 were 168, 168, 168, and 132, respectively. A summary of the charging and discharging phase operations is provided in Figure 6. Furthermore, Figure 7 illustrates the degradation trends of battery capacity across varying cycle numbers, which shows the capacity degrades below the threshold level of 70% for batteries $B005$,

Battery Name	CC-Charging		CV-Charging		CC-Discharging		Operating Condition	Initial Capacity	Cycle
	Constant current	Cut off voltage	Constant voltage	Cut off current	Constant current	Cut off voltage			
B005	1.5 A	4.2 V	4.2 V	20 mA	2 A	2.7 V	25 Celsius	1.86 Ah	168
B006	1.5 A	4.2 V	4.2 V	20 mA	2 A	2.5 V	25 Celsius	2.04 Ah	168
B007	1.5 A	4.2 V	4.2 V	20 mA	2 A	2.2 V	25 Celsius	1.89 Ah	168
B018	1.5 A	4.2 V	4.2 V	20 mA	2 A	2.5 V	25 Celsius	1.86 Ah	132

Fig. 6: Specifications of NASA AMES battery dataset.

$B006$, $B007$, and $B0018$ in 168, 168, 168, and 132 cycles. Additionally, to clarify the understanding of the dataset, we presented in Figure 8 the 2D- distribution of charge and discharge data for each battery.

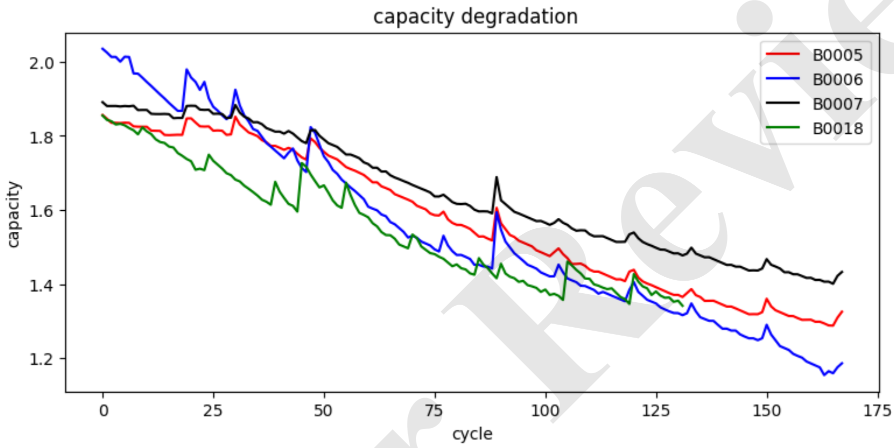
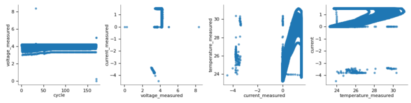


Fig. 7: Capacity degradation trends of NASA AMES battery dataset.

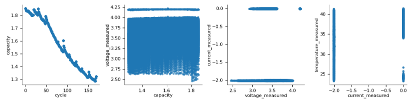
Finally, the sample of the charging and discharging data sets of one battery $B005$ are shown in Figures 9 and 10, respectively.

The proposed approach necessitates IC analysis for CC charging data. However, it's important to note that the battery's charging process comprises CC and CV phases. The CC-CV phase for each battery charge cycle is depicted in Figure 11. To extract health indicators (HIs) from the charging data, we specifically extract the constant current data from each cycle within the charge cycle data.

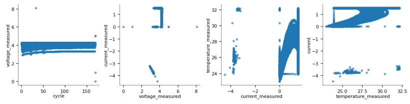
Figure 11 visually represents how current and voltage evolve during each phase of the battery charging cycle. It commences with the constant current (CC) phase, wherein the battery charges until it attains a predetermined voltage limit. Following this, the process seamlessly shifts into the constant voltage (CV) phase, where charging persists until the current descends to a specific threshold, typically set at 20 mA. Similarly, Figure 12 represents the voltage change concerning the discharge cycle for



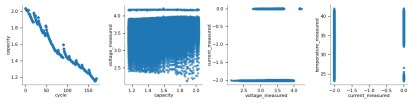
(a) For B005 battery charge cycle data



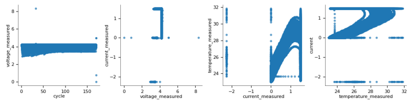
(b) For B005 battery discharge cycle data



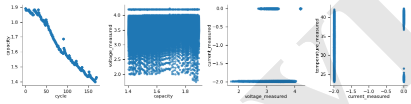
(c) For B006 battery charge cycle data



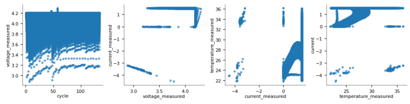
(d) For B006 battery discharge cycle data



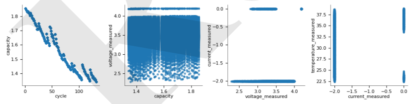
(e) For battery charge cycle data



(f) For B007 battery discharge cycle data



(g) For B018 battery charge cycle data



(h) For B018 battery discharge cycle data

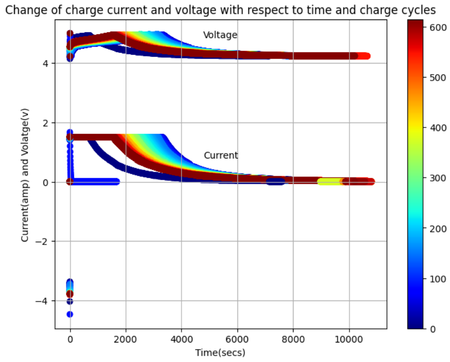
Fig. 8: 2D-data distribution for each NASA AMES battery dataset.

index	cycle	ambient_temperature	datetime	voltage_measured	current_measured	temperature_measured	current	voltage	time
0	1	24	2008-04-02 13:08:17	3.873017221300996	-0.001200660698297908	24.65535783391511	0.0	0.003	0.0
1	1	24	2008-04-02 13:08:17	3.479393559379272	-4.030268477538787	24.66647981177477	-4.036	1.57	2.532
2	1	24	2008-04-02 13:08:17	4.000587822429767	1.5127306474745377	24.675394470060205	1.5	4.726	5.5
3	1	24	2008-04-02 13:08:17	4.01239519378372	1.5090632819299827	24.693865093855624	1.5	4.742	8.343999999999998
4	1	24	2008-04-02 13:08:17	4.019708059348954	1.5113181929087844	24.705069460327003	1.5	4.753	11.125000000000004
5	1	24	2008-04-02 13:08:17	4.025409466688173	1.5127791303050797	24.718139716943565	1.498	4.758	13.891000000000002

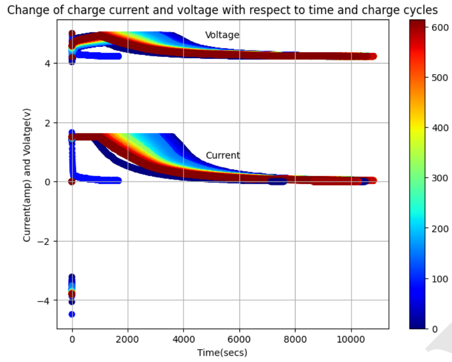
Fig. 9: Sample of charging dataset.

index	cycle	ambient_temperature	datetime	capacity	voltage_measured	current_measured	temperature_measured	current	voltage	time
0	1	24	2008-04-02 15:25:41	1.8564874208181574	4.191491807505295	-0.004901589207462691	24.330033885570543	-0.0006	0.0	0.0
1	1	24	2008-04-02 15:25:41	1.8564874208181574	4.190749067776103	-0.0014780055516425076	24.325993424022467	-0.0006	4.206	16.781
2	1	24	2008-04-02 15:25:41	1.8564874208181574	3.9748709122299895	-2.0125283240860368	24.389085127564876	-1.9982	3.062	35.702999999999996
3	1	24	2008-04-02 15:25:41	1.8564874208181574	3.9517167075108937	-2.0139793620987403	24.544752316727223	-1.9982	3.03	53.781
4	1	24	2008-04-02 15:25:41	1.8564874208181574	3.934352489432596	-2.011143595991146	24.731385200934106	-1.9982	3.011	71.922
5	1	24	2008-04-02 15:25:41	1.8564874208181574	3.920058441667911	-2.0130066278129433	24.909816397303786	-1.9982	2.991	90.094
6	1	24	2008-04-02 15:25:41	1.8564874208181574	3.9079035099130923	-2.0143996570526377	25.105884152527363	-1.9982	2.977	108.281

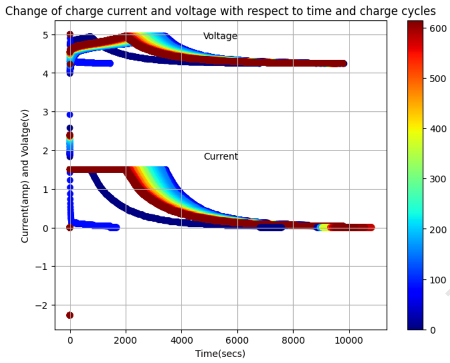
Fig. 10: Sample of discharging dataset.



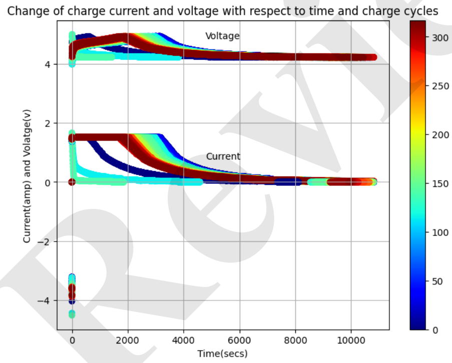
(a) Battery B005



(b) Battery B006



(c) Battery B007



(d) Battery B018

Fig. 11: Current and voltage of charging NASA AMES battery dataset in different charge cycles.

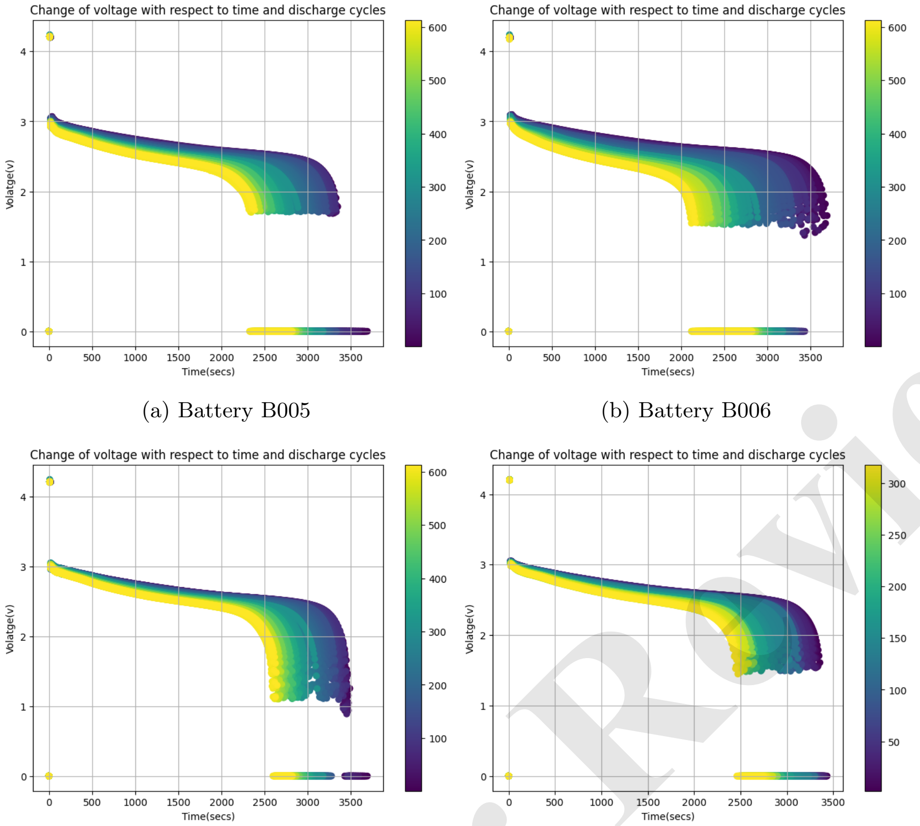
each battery. Figure 12 shows that the voltage is dropped out as the discharge cycle progresses.

4.3 Multi health indicators extraction

This section explores the methodology used to calculate the health indicators used in this paper. We computed health indicators for each of the four batteries.

4.3.1 ICA based HIs extraction

In recent years, the estimation of the RUL of lithium-ion batteries (LIB) using the incremental capacity analysis (ICA) method has gained significant attention from scholars. ICA transforms the low-dimensional voltage-capacity curve of the LIB (referred to as the V-Q curve) into a higher-dimensional IC curve. This IC curve,



(a) Battery B005 (b) Battery B006

(c) Battery B007 (d) Battery B018

Fig. 12: Change in voltage of NASA AMES battery dataset in different discharge cycles.

represented as the dQ/dV - V curve, is derived by calculating the difference between battery capacity changes and battery terminal voltage changes. The formulation for computing the IC curve is as follows:

$$IC_{charging} = \frac{dQ_c}{dV_c} \approx \frac{\Delta Q_c}{\Delta V_c} = \frac{i_c \Delta t}{V_{c,2} - V_{c,1}} \tag{3}$$

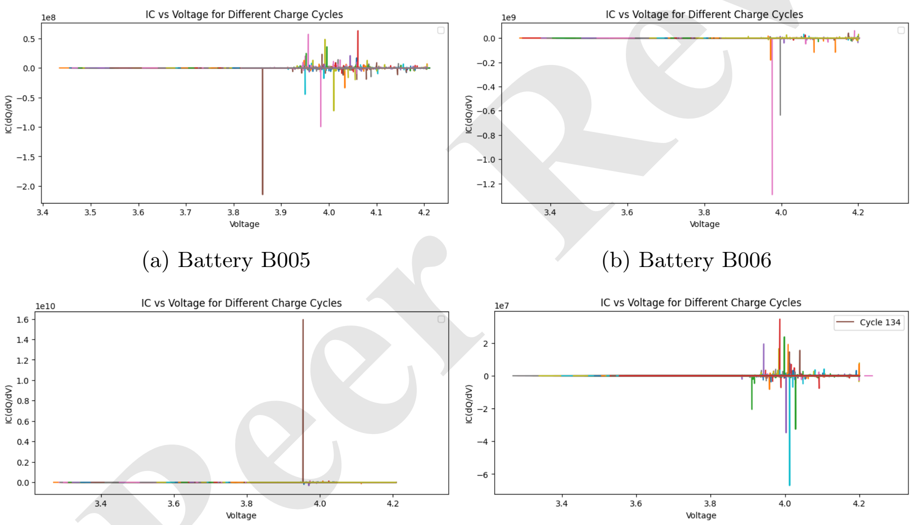
$$IC_{discharging} = \frac{dQ_d}{dV_d} \approx \frac{\Delta Q_d}{\Delta V_d} = \frac{i_d \Delta t}{V_{d,2} - V_{d,1}} \tag{4}$$

Q_c signifies the battery's charging capacity, while V_c represents its terminal voltage during the charging phase. On the other hand, Q_d denotes the battery's discharging capacity, and V_d corresponds to its terminal voltage during the discharging process.

Furthermore, we have i_c, which denotes the charging current, and i_d, representing the discharging current. The variable t is employed to represent time.

The IC curves method effectively translates the voltage plateaus, characteristic of the battery's behaviour, into identifiable variations on the IC curve. These variations are instrumental for accurately detecting subtle changes as the battery ages [60, 61]. Notably, the extent of reactions between the battery's electrodes is influenced by the aging process, particularly during charging. While these variations may not be readily apparent in the voltage curve, they manifest as inflection points or peaks in the IC curve [61]. This provides a valuable means to monitor the battery's condition and potentially forecast its aging trajectory. As capacity slowly degrades due to battery aging, these changes are quantitatively identifiable within the IC curve, making it a powerful tool for assessing battery health and performance.

Consequently, within the context of our analysis, the height and position of the peak within the CC charging IC curve serve as the basis for deriving two health indices, HI1 and HI2. Similarly, the height and position of the peak within the discharging IC curve are utilized to compute our other two health indices, HI3 and HI4. The IC curves, crucial for our analysis, are depicted in Figures 13 and 14 for each battery across charge and discharge cycles, respectively.

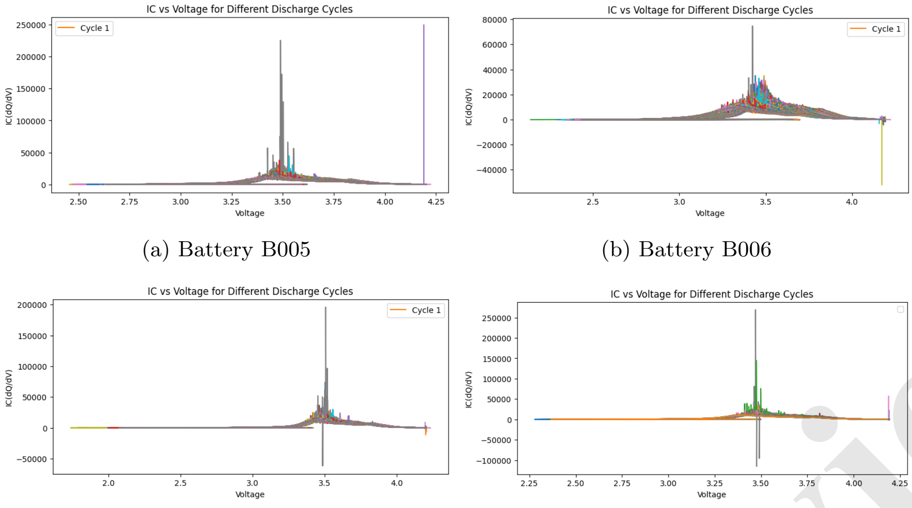


(a) Battery B005 (b) Battery B006

(c) Battery B007 (d) Battery B018

Fig. 13: IC curve in CC-charge cycles.

In Figure 13, the IC curve is depicted, which was derived from the CC charge cycle data for cycle 1. Meanwhile, Figure 14 illustrates the IC curve obtained from the CC discharge cycle.



(a) Battery B005 (b) Battery B006

(c) Battery B007 (d) Battery B018

Fig. 14: IC curve in CC-discharge cycles.

4.3.2 Direct Analysis

We have incorporated capacity and cycle data directly from the dataset as two additional health indicators (HIs). In conjunction with the existing four HIs computed with the IC curve, these form six HIs employed in predicting the RUL of EV lithium-ion batteries. The sample of these HIs is visually presented in Figure 15. Where the

cycle	capacity	IC_C_H	IC_C_P	IC_D_H	IC_D_P
1	1.856487	88289.81	4.206847	22830.11	3.480996
2	1.846327	59498.18	3.971994	24381.93	3.501615
3	1.835349	64157.53	4.210378	24668.76	3.484151
4	1.835263	85564.97	3.998606	25125.65	3.497942
5	1.834646	511873.2	3.977384	21963.46	3.512115

Fig. 15: Sample screenshot of HIs.

terms IC_C_H, IC_C_P, IC_D_H, and IC_D_P represent the extracted HIs associated with the height and position of the peaks in the charge and discharge IC curves, respectively.

4.4 Correlation analysis

The correlation coefficient is a statistical metric quantifying the degree of a linear association between two variables. It spans a scale from -1 to 1. A correlation coefficient of -1 signifies a strongly negative or inverse correlation, where one series of values increases as the other decreases, and conversely. We have performed a correlation analysis of battery capacity's health indices (HIs). We aimed to learn more about the relationships among these computed HIs and determine their collective capacity to reflect changes in battery capacity. To achieve this, we employed Pearson and Spearman correlation analyses to examine the interactions between the HIs and battery capacity.

Pearson correlation analyses mathematically expressed as follows:

$$\rho = \frac{\sum (E - \bar{E})(F - \bar{F})}{\sqrt{\sum (E - \bar{E})^2 \sum (F - \bar{F})^2}} \tag{5}$$

Where ρ is the correlation coefficient, n is the number of samples, E is the eigenvalue, F is battery SOH, and $\bar{E}$ and $\bar{F}$are the corresponding average values. The judgement method is that the larger the value of $|\rho|$, the higher the degree of correlation.

Additionally, Spearman correlation analyses are mathematically expressed as follows:

$$\rho = 1 - \frac{6 \sum d_i^2}{n(n^2 - 1)} \tag{6}$$

Here, ρ represents the Spearman's rank correlation coefficient. The variable d_i is characterized as the disparity between the rankings of each observation, with n representing the total count of observations.

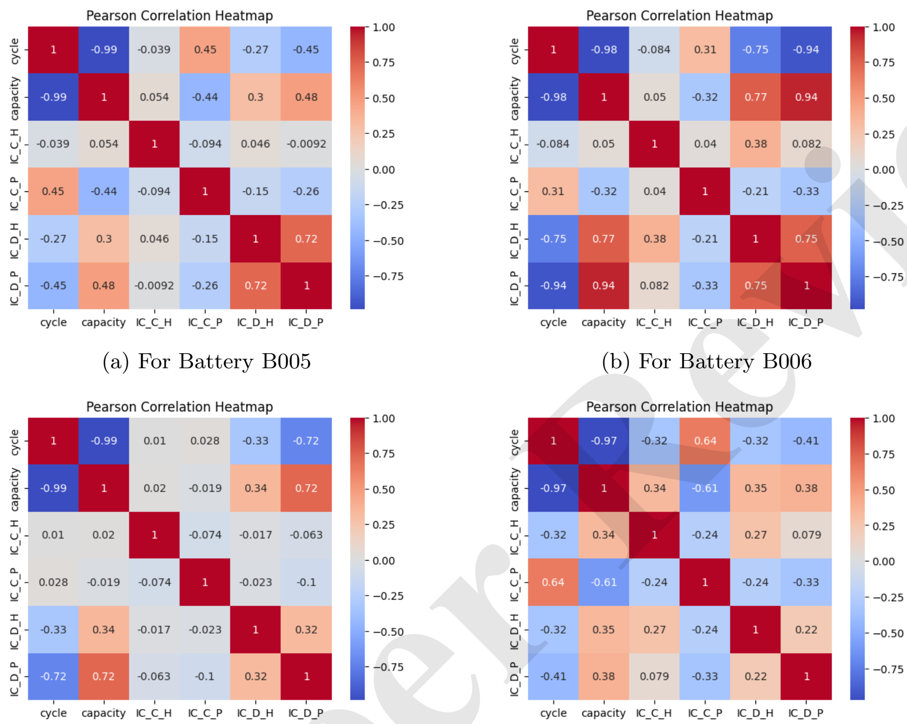
The Pearson and Spearman correlation analyses between the HIs and the battery's capacity are depicted in Tables 2 and 3. The data in these tables shows significant correlations between the HIs and the battery's capacity, especially with health indicating HI3 and HI4, obtained from the discharge cycle Ic curve. It indicates that the health indicator obtained from the discharge cycle IC curve represents a reliable indicator of capacity changes. We have also presented the heatmap of Pearson and Spearman correlation analyses for each battery in Figures 16 and 17. The heatmap shows the correlations among the HIs. Furthermore, we have also shown in Figure 18 the distribution of relationships between our discharge cycle HIs and capacity. It depicts the strong positive correlation between HI3 and HI4 with the capacity.

Table 2: Pearson correlation analysis of each HIs with capacity.

Battery Name	**HI1 (IC_C_H)**	**HI2 (IC_C_P)**	**HI3 (IC_D_H)**	**HI4 (IC_D_P)**	**Cycles**
B005	0.054	-0.44	0.3	0.48	-0.99
B006	0.05	-0.32	0.77	0.94	-0.98
B007	0.02	-0.019	0.34	0.72	-0.99
B0018	0.34	-0.61	0.35	0.38	-0.97

Table 3: Spearman correlation analysis of each HIs with capacity..

Battery Name	HI1 (IC_C_H)	HI2 (IC_C_P)	HI3 (IC_D_H)	HI4 (IC_D_P)	Cycles
B005	-0.093	-0.51	0.88	0.83	-0.99
B006	0.07	-0.24	0.94	0.94	-0.99
B007	-0.24	-0.22	0.83	0.76	-0.99
B0018	0.59	-0.61	0.95	0.84	-0.97



(a) For Battery B005

(b) For Battery B006

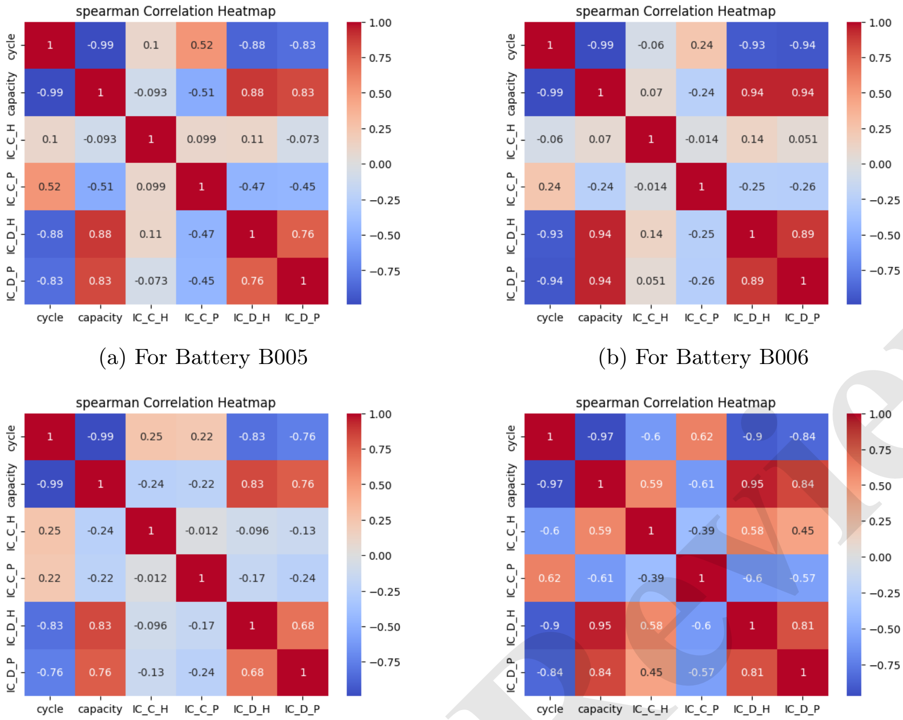
(c) For Battery B007

(d) For Battery B018

Fig. 16: Pearsion correlations matrix between the HIs.

4.5 Training process

Initially, we extract the six HIs using the IC curve and direct analysis. Subsequently, we employ a deep learning model called Stacked LSTM (SLSTM) for training and testing with these HIs to estimate RUL. This proposed model is referred to as $S{-}LSTM{+}HI$. We have also developed and implemented additional models for comparative purposes,



(a) For Battery B005

(b) For Battery B006

(c) For Battery B007

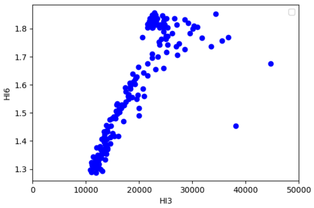
(d) For Battery B0018

Fig. 17: Spearman correlations matrix between the HIs.

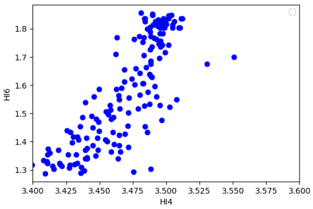
including Attention-Stacked LSTM (ASLSTM)+HI, GRU+HI, and AGRU+HI. A detailed list of parameters for these models can be found in Table 4.

Table 4: List of model parameters.

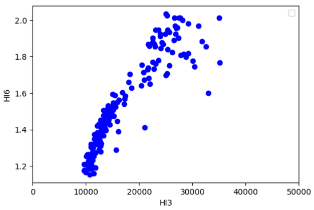
S.No.	**Parameters**	**Values**
1	Epoch	200
2	Learning rate	0.1
3	optimizer	SGD
4	Activation	Relu
5	Loss	Mean squared error
6	Batch size	1
7	Layer structure	(60, 64, 32, 1)



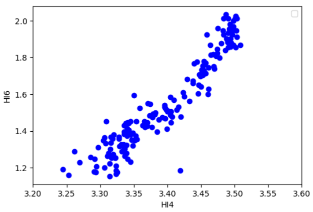
(a) For Battery B005



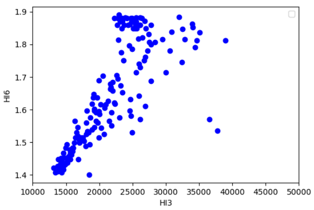
(b) For Battery B005



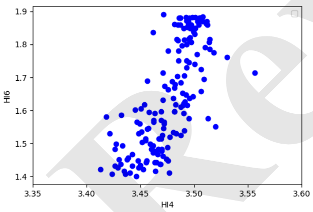
(c) For Battery B006



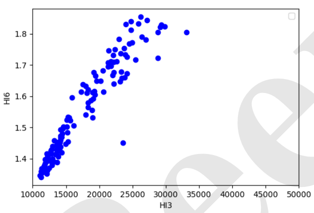
(d) For Battery B006



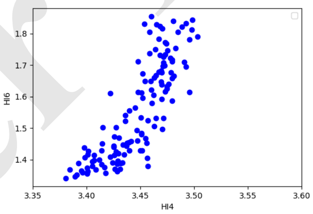
(e) For Battery B007



(f) For Battery B007



(g) For Battery B018



(h) For Battery B018

Fig. 18: Relationship of HI3 and HI4 with capacity of each battery.

We have employed each of the aforementioned models in three distinct scenarios [24, 25]:

i. **Case-1 (individual battery):** We divided each battery's HIs into training and testing sets using an 80:20 ratio. Specifically, we employed the subsequent training-test combinations:
   - Train with HI's of $B005$; predict for $B005$.

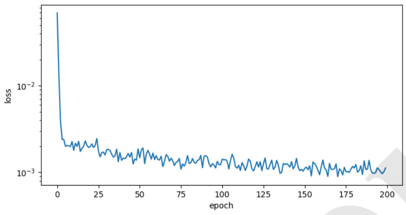
- Train with HI's of $B006$, predict for $B006$.
- Train with HI's of $B007$, predict for $B007$.
- Train with HI's of $B018$; predict for $B018$.

ii. **Case-2 (2P1):** We used the HIs data from two batteries for training to predict the capacity of another battery. Specifically, we employed the subsequent training-test combinations:
- Train with HI's of $B005$ and $B006$, predict for $B007$.
- Train with HI's of $B005$ and $B006$, predict for $B018$.
- Train with HI's of $B007$ and $B006$, predict for $B005$.
- Train with HI's of $B007$ and $B018$, predict for $B006$.

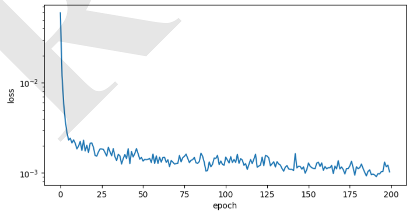
Each combination allowed us to evaluate the model's predictive performance on a different target battery.

iii. **Case-3 (leave-one-out):** We used the HI data from three batteries for training to predict the capacity of another battery. Specifically, we employed the subsequent training-test combinations:
- Train with HI's of $B006$, $B007$, and $B018$, predict for $B005$.
- Train with HI's of $B005$, $B007$, and $B018$, predict for $B006$.
- Train with HI's of $B005$, $B006$, and $B018$, predict for $B007$.
- Train with HI's of $B005$, $B006$, and $B007$, predict for $B018$.

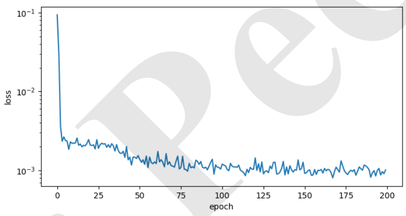
The training loss results for the proposed model are presented in Figure 19, Figure 20, and Figure 21. Each figure demonstrates the effective convergence of the proposed model within 200 epochs.



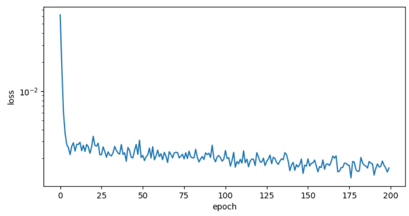
(a) For Battery B005



(b) For Battery B006

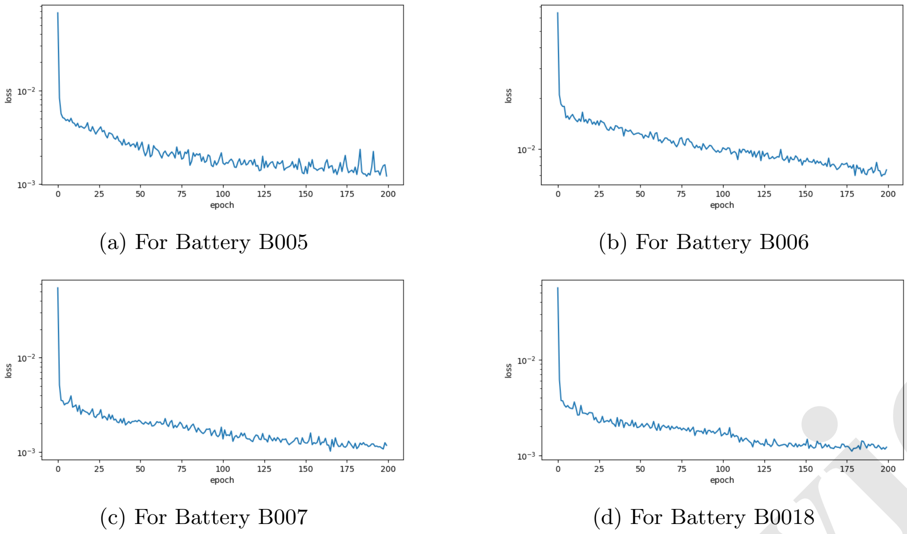


(c) For Battery B007



(d) For Battery B0018

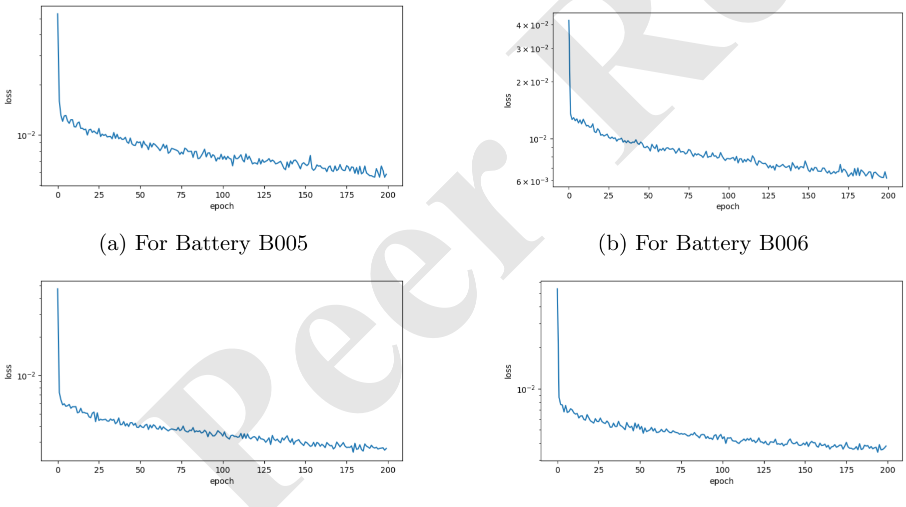
Fig. 19: Training loss for individual batteries.



(a) For Battery B005 (b) For Battery B006

(c) For Battery B007 (d) For Battery B0018

Fig. 20: Training loss of batteries for 2P1 strategy.



(a) For Battery B005 (b) For Battery B006

(c) For Battery B007 (d) For Battery B0018

Fig. 21: Training loss of batteries for leave-one-out strategy.

5 Results

This section describes the experimental platform, performance measures, and results obtained from our experiments.

5.1 Experimental platform

The proposed models $SLSTM + HI$ are evaluated against models ASLSTM+HI, GRU+HI, and AGRU+HI. All the methods were implemented using Python 3.8.1 and Keras 2.13.1, an open-source deep learning library built on TensorFlow 2.13.0. The experiments were conducted on the Google Colab platform with an NVidea K80, 12 GB GPU, and a 2.5 GHz Intel (R) Xeon (R) processor with 256 GB of storage.

5.2 Performance measures

To assess the impact of the proposed model $SLSTM + HI$ in EV battery RUL prediction, we use three widely used regression prediction indicators—the root-mean-square error (RMSE), the mean absolute error (MAE), and the R-square (R^2) as the evaluation criteria of the methods. Details of these measures are provided below:

1. **RMSE:** It measures the average deviation between a model's predicted values and the values actually found in a dataset. The procedure for calculating RMSE is as follows:

$$\mathrm{RMSE}(C, \hat{C}) = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(C - \hat{C})^2} \tag{7}$$

   where $\hat{C}$ denotes the predicted value of capacity, and C denotes the real value of capacity. The lower the RMSE, the more precise the prediction.

2. **MAE:** It is yet another popular metric for assessing the performance of error detection models or algorithms. MAE calculates the average absolute difference between predicted and actual values in a dataset. MAE is calculated as follows:

$$\mathrm{MAE}(C, \hat{C}) = \frac{1}{n}\sum_{i=1}^{n}\left|\hat{C} - C\right| \tag{8}$$

   where $\hat{C}$ denotes the predicted value of capacity, and C denotes the real value of capacity. The lower the RMSE, the more precise the prediction

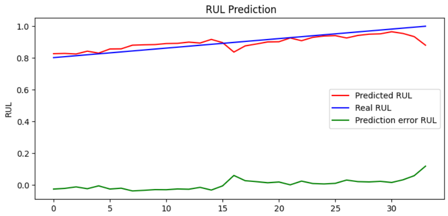
3. **R-square (R^2):** This measurement is commonly referred to as the coefficient of determination, indicating how effectively a model aligns with the data. In simpler terms, it serves as a statistical measure of how well the regression line fits the actual data within the regression context. The calculation for R^2 is as follows:

$$R^2(C, \hat{C}) = 1 - \frac{\sum_{i=1}^{N}(C - \hat{C})^2}{\sum_{i=1}^{N}(\bar{C} - \hat{C})^2} \tag{9}$$

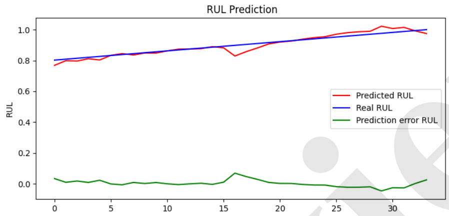
   where $\hat{C}$ denotes the predicted value of capacity and C denotes the actual value of capacity. R^2 has a value between 0 and 1. The nearer it is to one, the higher the model's performance.

Table 5: Comparative analysis of the proposed model with variants in individual battery strategy.

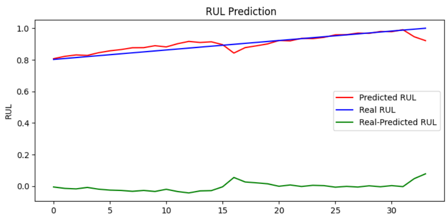
Batteries	Proposed (SLSTM+HI)		A-SLSTM+HI		GRU+HI		AGRU+HI	
	MAE	RMSE	MAE	RMSE	MAE	RMSE	MAE	RMSE
B005	0.0259	0.0331	0.0265	0.0337	0.0267	0.0331	0.0305	0.0392
B006	0.0156	0.0220	0.0116	0.0182	0.0146	0.0197	0.0690	0.0774
B007	0.0192	0.0261	0.0276	0.0272	0.0200	0.0265	0.0461	0.0560
B018	0.1052	0.1124	0.0887	0.0956	0.0672	0.0772	0.0615	0.0707



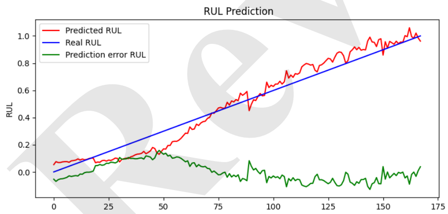
(a) For Battery B005



(b) For Battery B006



(c) For Battery B007

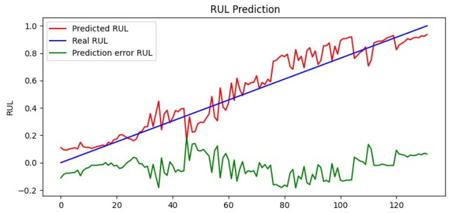


(d) For Battery B0018

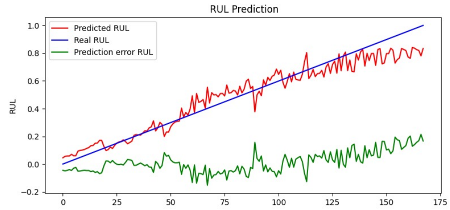
Fig. 22: RUL prediction and error plots for individual batteries using the proposed model.

5.3 Results in case-1:

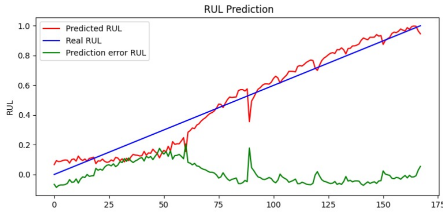
The results for Case-1, where the HIs of each battery were divided into training and testing sets using an 80:20 ratio, are presented in Table 5. As depicted in Table 5, it's evident that the MAE and RMSE of the proposed model for batteries B005 and B007 are notably low at 0.0259 and 0.0331, as well as 0.0192 and 0.0261 respectively, surpassing the performance of the other models shown. Moreover, the ALSTM+HI model demonstrates superior predictive accuracy for battery $B006$, while the AGRU+HI model excels in predicting the RUL of battery B018. To provide additional insight into the performance of the proposed model in predicting RUL, we have included error plots alongside the RUL predictions for the test data in Figure 22 when the proposed model is applied.



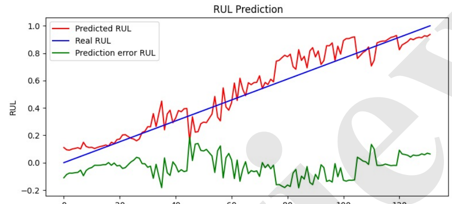
(a) For Battery B005



(b) For Battery B006



(c) For Battery B007



(d) For Battery B0018

Fig. 23: RUL prediction and error plots for 2P1 case using the proposed model.

5.4 Results in case-2:

In Case-2 (2P1), we utilized HIs data from two batteries for training to predict the capacity of another battery. The results of the proposed model in forecasting remaining Useful Life and the corresponding error plots for the test data are presented in Figure 23.

5.5 Results in case-3:

In Case-3 (Leave-one-out), we employed HI data from three batteries for training to forecast the capacity of another battery. The results for Case-3 are presented in Table 6. As demonstrated in the table, the MAE, RMSE, and R^2-Score of the proposed model for batteries B005, B006, B007, and B018 are impressively low at 0.0589, 0.0730, and 0.9357 respectively. These outcomes underscore the superior performance of the proposed model in predicting the remaining useful life of EV lithium-ion batteries compared to other models.

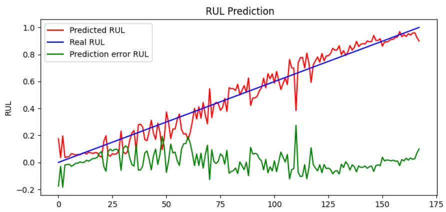
To further illustrate the predictive capability of the proposed model for RUL, we have included error plots alongside the predictions for the test data in Figure 24 when the proposed model is applied.

5.6 Comparative results with related works

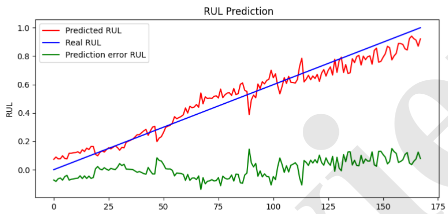
We performed a comparative analysis of our models against the relevant existing methods outlined below:

Table 6: Comparative analysis of the proposed model with variants in Leave-one-out strategy.

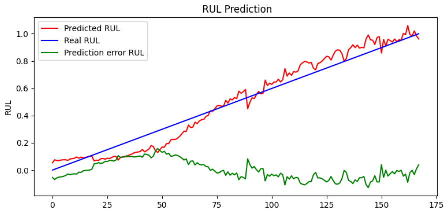
Batteries	**Proposed (SLSTM+HI)**			**ASLSTM+HI**			**GRU+HI**			**AGRU+HI**		
	MAE	**RMSE**	R^2 **Score**	**MAE**	**RMSE**	R^2 **Score**	**MAE**	**RMSE**	R^2 **Score**	**MAE**	**RMSE**	R^2 **Score**
B005	0.0546	0.0704	0.9412	0.0659	0.0848	0.9147	0.0613	0.0774	0.9290	0.0719	0.0836	0.9171
B006	0.0532	0.0634	0.9524	0.0619	0.0679	0.9454	0.0685	0.0789	0.9262	0.0545	0.0712	0.9399
B007	0.0583	0.0689	0.9438	0.0567	0.0716	0.9392	0.0676	0.0805	0.9232	0.0698	0.0838	0.9167
B018	0.0693	0.0894	0.9055	0.0764	0.0952	0.8930	0.0783	0.0974	0.8878	0.0848	0.0977	0.8872
AVG	**0.0589**	**0.0730**	**0.9357**	0.0652	0.0799	0.9231	0.0689	0.0836	0.9166	0.0703	0.0841	0.9152



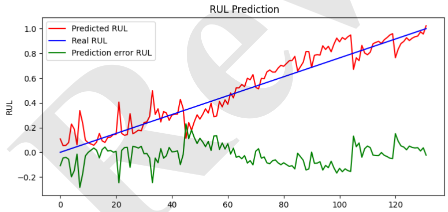
(a) For Battery B005



(b) For Battery B006



(c) For Battery B007



(d) For Battery B0018

Fig. 24: RUL prediction and error plots for leave-one-out case using the proposed model.

- RNN-10 [21]: In this work, With the inclusion of multiple Recurrent Neural Network (RNN) units, the primary aim is to predict the RUL.
- GRU-18 [23]: This work incorporates a GRU unit for learning and capturing the degradation trend from the input sequence.
- Dual-LSTM-19 [22]: In this work, employing two distinct LSTM cell types enables the capture of non-linearity in the relationship between capacities.
- MLP-19 [20]: In this study, the utilization of multiple fully connected layers serves as a means to effectively capture the dynamic and nonlinear degradation trends in battery performance.
- LSTM-22 [8]: This work incorporates multiple Long Short-Term Memory (LSTM) units. The intention is to leverage their capabilities in learning and capturing the degradation trend of LIBs.

Table 7: Comparative analysis of the proposed work with related works in Leave-one-out strategy.

	RNN-10 [21]	GRU-18 [23]	D-LSTM-19 [22]	MLP-19 [20]	LSTM-22 [8]	DE-Transformer-22[24]	AutoML-22 [32]	Proposed-SLSTM+HI
MAE	0.0749	0.0806	0.0815	0.1379	0.0829	0.0713	0.0336	0.0589
RMSE	0.0848	0.0921	0.0879	0.1541	0.0905	0.0802	0.0414	0.0730
%MAE Improved	21.36 %	26.92 %	27.73 %	57.29 %	28.95 %	17.39 %	-42.95 %	-
%RMSE Improved	13.92 %	20.74 %	16.95 %	52.63 %	19.34 %	8.98 %	-43.29 %	-

- DeTransformer-22 [24]: In this research, a denoising autoencoder was employed to extract meaningful representations from corrupted battery charge/discharge data. Subsequently, these representations were utilized to reconstruct the original input. Following this, Transformer networks were applied to predict the RUL of the LIBs.
- AutoML-22 [32]: In this work, three HIs are extracted during the Constant Current-Constant Voltage (CC-CV) charging phase. These HISs are then employed as inputs for an AutoML prediction model, utilized for the RUL prediction of Lithium-ion Batteries.

To validate the performance of the proposed model, we conducted a comparative analysis with the related works mentioned earlier using the leave-one-out approach, similar to the most recent work [32]. Our comparative results are depicted in Table 7. Table 7 clearly illustrates that the proposed approach outperforms all other works regarding MAE and RMSE performance measures. Furthermore, the proposed methods show significant improvements in MAE compared to existing models: RNN-10 [21] by 21.36 % $[((0.0749-0.0589)/0.0749)\times 100]$, GRU-18 [23] by 26.92 %, D-LSTM-19 [22] by 27.73 %, MLP-19 [20] by 57.29 %, LSTM-22 [8] by 28.95 %, and DE-Transformer-22 [24] by 17.39 %. Similarly, RMSE is improved by 13.92 %, 20.74 %, 16.95 %, 52.63 %, 19.34 %, and 8.98 %, respectively.Moreover, the proposed methods demonstrate performance comparable to the most recent work [32], which achieved 42.95 % better MAE and 43.29 % better RMSE using the leave-one-out strategy.

Finally, we conducted a comparison between the results from our proposed model and the latest research presented in [32] for case-1 in Table 8. This comparison revealed that in case 1, which involves individual battery analysis, our proposed model excels in predicting the RUL for batteries B005, B006, and B007, except for battery B018. Consequently, our proposed model demonstrates superior RUL prediction performance for electric vehicle lithium-ion batteries compared to the related existing works [8, 20–24]. It also exhibits performance comparable to the most recent research in [32]. While the proposed model performs better for individual batteries, the work presented in [32] excels in the leave-one-out strategy.

Table 8: Comparative analysis of the proposed work with AutoML-22 [32] for individual batteries.

Batteries	Proposed (SLSTM+HI)		AutoML-22 [32]	
	MAE	RMSE	MAE	RMSE
B005	0.0259	0.0331	0.0283	0.0337
B006	0.0156	0.0220	0.0221	0.0340
B007	0.0192	0.0261	0.0361	0.0407
B018	0.01052	0.1124	0.0479	0.0573
AVG	0.0178	0.0484	0.0336	0.0414

6 Conclusion

This paper presents a remaining useful life (RUL) prediction model for lithium-ion batteries (LIBs) using stacked long short-term memory (SLSTM) deep learning models. The model incorporates novel Pearson and Spearman correlation-based multi-health indicators (HIs) extracted using the peaks of the IC curve from the constant current charging (CC) and discharging phases on battery aging. Specifically, the SLSTM model uses six HIs: two from the CC charging phase, two from the CC discharging phase, and two representing cycle and capacity information obtained through direct analysis. To evaluate the effectiveness of our proposed model in predicting the RUL of LIBs, we developed and implemented additional benchmark models for comparative analysis, including ASLSTM+HI, GRU+HI, and AGRU+HI. A comparative evaluation of these benchmark models and recent related models on NASA's LIB battery dataset reveals that the proposed model outperforms existing works in individual batteries, 2P1, and leave-one-out scenarios. Furthermore, our model accurately predicts the RUL of LIBs, achieving a mean absolute error (MAE) of 0.0589 and a root mean square error (RMSE) of 0.0730.

Acknowledgements

Authors acknowledge Visvesvaraya PhD Scheme, MeitY, Govt. of India MEITY-PHD-2525 for supporting this research.

Statements and Declarations

- Funding: This work did not receive financial support.
- Conflict of interest/Competing interests: The authors declare that there is no conflict of interest regarding the publication of this paper.
- Ethics approval: Not applicable
- Consent to participate: All the authors declare their consent to participate in this research article.
- Consent for publication: All the authors declare their consent for publication of the article on acceptance.
- Availability of data and materials: The data and materials are available within the manuscript.
- Code availability: Not applicable

- Authors' contributions: Shivendu Mishra: Conceptualization, Investigation, Methodology, Writing—original draft. Anurag Choubey: Conceptualization, Investigation, Methodology, Writing—review & editing. Bollampalli Areen Reddy: Designing, Implementing. Rajiv Misra: Validation, Writing—review & editing, Supervision.

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